Options: a quick recap

Quantitative Finance

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What is an option?

An **option** is a contract between a *buyer* (holder) and a *seller* (writer). It gives the buyer a *right* - but *not* an obligation.

Call option

the right to *buy* the underlying asset at a fixed *strike price* K, at (or up to) the *maturity* T.

Put option

the right to *sell* the underlying asset at a fixed strike price K, at (or up to) maturity T.

- the buyer pays a *premium* (the option price) at inception - a right has positive value;
- the writer has the matching *obligation* (to sell, for a call; to buy, for a put) whenever the buyer exercises.

Payoff at maturity

We focus on *European-style* options: exercise only at T. The payoff then depends on the terminal price S_T only; we write it generically as $C_T = g(S_T)$.

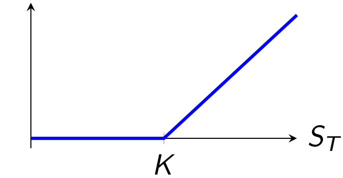
- call, strike K: payoff $\max\{S_T - K, 0\}$
- put, strike K: payoff $\max\{K - S_T, 0\}$
- digital (binary) call: payoff $1\{S_T \geq K\}$
- digital (binary) put: payoff $1\{S_T \leq K\}$

(the indicator $1\{A\}$ equals 1 if A is true and 0 otherwise.)

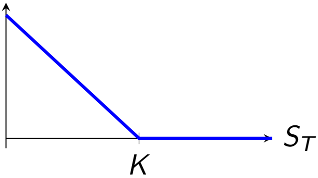
The *writer*'s payoff is the buyer's payoff with a minus sign: e.g. writing a call gives $-\max\{S_T - K, 0\} \leq 0$ at maturity.

The payoff $C_T = g(S_T)$ of each contract, as a function of the stock price S_T:

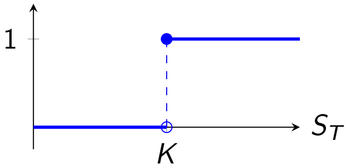
call: $\max\{S_T - K, 0\}$



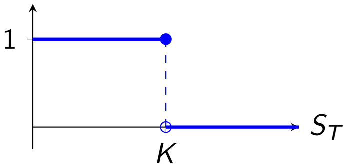
put: $\max\{K - S_T, 0\}$



digital call: $1\{S_T \geq K\}$



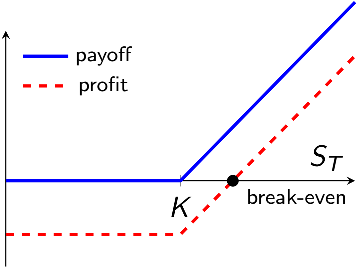
digital put: $1\{S_T \leq K\}$



Payoff versus profit

The *payoff* ignores the premium paid up front; the *profit* subtracts it. For a long call (strike K, premium c paid at $t = 0$):

- payoff $= \max\{S_T - K, 0\} \geq 0$
- profit $= \max\{S_T - K, 0\} - c\ e^{rT}$
- break-even at $S_T = K + c\ e^{rT}$
- max. loss $= c\ e^{rT}$ (the premium); upside unlimited



Moneyness

Describes where the strike sits relative to the current price S_t:

	Call	**Put**
in-the-money (ITM)	$S_t > K$	$S_t < K$
at-the-money (ATM)	$S_t \approx K$	$S_t \approx K$
out-of-the-money (OTM)	$S_t < K$	$S_t > K$

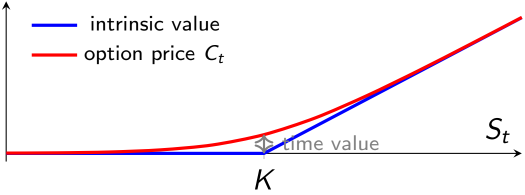
A European option is exercised at T exactly when it is ITM at T; otherwise it expires worthless. "Moneyness" is a convenient shorthand when talking about an option's current state.

Time value of an option

Before maturity the option price splits into two parts (shown for a call):

$$\underbrace{C_t}_{\text{price}} = \underbrace{\max\{S_t - K, 0\}}_{\text{intrinsic value}} + \underbrace{(\text{time value})}_{\geq 0}.$$

Intrinsic value is the payoff if the option were exercised now; *time value* is the extra worth from the chance of moving (further) into the money before T. It decays to 0 as $t \to T$ ("time decay"); at maturity C_T equals the intrinsic value (why ≥ 0? see next slide).



Remark. We will *not* use this intrinsic/time-value split in the remainder of the course - we always work with the full price C_t produced by a model.

Why is the time value of a call nonnegative?

Consider a European call on a stock that pays no dividends; let $r \geq 0$ be the risk-free rate. Two no-arbitrage observations:

1. $C_t \geq 0$: the payoff $\max\{S_T - K, 0\}$ is never negative, so the call cannot trade at a negative price.
2. $C_t \geq S_t - K\,\mathrm{e}^{-r(T-t)}$: the call plus $K\,\mathrm{e}^{-r(T-t)}$ in the bank pays $\max\{S_T - K, 0\} + K = \max\{S_T, K\} \geq S_T$ at maturity, i.e. at least as much as one share, in *every* scenario; hence $C_t + K\,\mathrm{e}^{-r(T-t)} \geq S_t$.

Combining the two bounds, and using $K\,\mathrm{e}^{-r(T-t)} \leq K$ (as $r \geq 0$),

$$C_t \;\geq\; \max\{S_t - K\,\mathrm{e}^{-r(T-t)},\, 0\} \;\geq\; \max\{S_t - K,\, 0\},$$

so the price is at least the intrinsic value: the time value is nonnegative.

Food for thought

Does the same conclusion hold for a European *put*? Mimic the argument: which lower bound on P_t do you get, and does it dominate the intrinsic value $\max\{K - S_t, 0\}$? Consider a deep in-the-money put ($S_t \approx 0$) with $r > 0$.

Exercise styles

When may the holder exercise?

- *European*: only at the expiration date T;
- *American*: at any time up to and including T;
- there are also *Asian*, *Bermudan*, *Canary*, . . . variants (the names have nothing to do with geography).

- rule of thumb: options on indices and interest rates are typically European; options on individual stocks and ETFs are typically American.
- **This course focuses on European-style options** (on stocks, indices, ETFs). American-style options and interest-rate options are treated in the MSc QFAS.

Where are options traded, and some terminology

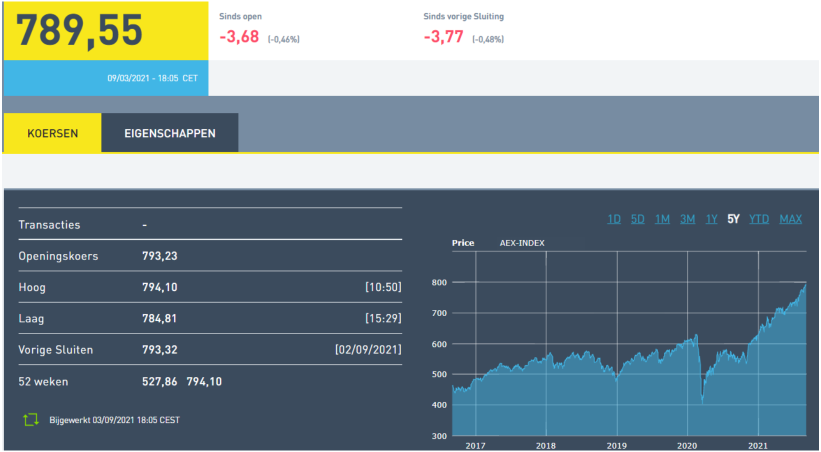
- options trade on *exchanges*, in *over-the-counter* (OTC) markets, and as bespoke contracts written by investment banks;
- the *underlying* can be a stock, an index, an ETF, an interest rate, real estate, ...;
- an option is an example of a *financial derivative*: a contract whose value is defined in terms of an underlying asset traded on a financial market.

Terminology used interchangeably:

- *strike price* = *exercise price* K;
- *maturity* = *expiration date* T (the final date for exercising).

Example: the AEX index

As a running example we use European call and put options on the AEX index. (Illustrative market data, 3 September 2021; the precise figures are not important.)



Example: an AEX option chain

Quoted call and put prices on the AEX index (3 September 2021), expiring December 2022. Note how prices vary with the strike K.

AEX-INDEX

EURONEXT DERIVATIVES AMSTERDAM
INDEX OPTION

AEX	Aantal 03/09/2021 17:29	Openstaande positie 02/09/2021	Valuta	Tijdzone
	24 732	287 683	EUR	CET

DECEMBER 2022 KOERSEN - 03/09/21

	SETTL.	LAATSTE	BIED	LAAT		UITOEFENPRIJS		BIED	LAATSTE	SETTL.
+	230.19	-	225.25	238.95	C	**550.00**	P	13.60	14.10	14.03
+	185.35	185.00	183.00	188.50	C	**600.00**	P	19.10	18.90	19.56
+	142.83	-	141.00	144.00	C	**650.00**	P	26.80	27.20	27.40
+	103.35	100.00	102.00	104.00	C	**700.00**	P	37.50	38.20	38.29
+	68.30	68.20	67.60	68.90	C	**750.00**	P	52.70	-	53.62
+	40.26	41.55	39.50	40.70	C	**800.00**	P	74.10	76.80	75.94
+	20.67	20.45	20.30	20.90	C	**850.00**	P	-	106.50	106.72
+	9.39	9.50	9.05	9.65	C	**900.00**	P	140.00	-	145.81
+	4.16	3.90	3.85	4.40	C	**950.00**	P	-	-	190.95
+	1.89	2.00	1.70	2.15	C	**1000.00**	P	-	-	239.05
+	0.53	0.63	0.25	0.80	C	**1100.00**	P	-	-	338.42

Quiz

At maturity the stock is worth $S_T = 42$. For strike $K = 40$, give the payoff of:

1. a (long) call;
2. a (long) put;
3. a digital call;
4. the position of the *writer* of the call.